

END-TO-END E-COMMERCE INTELLIGENCE SYSTEM

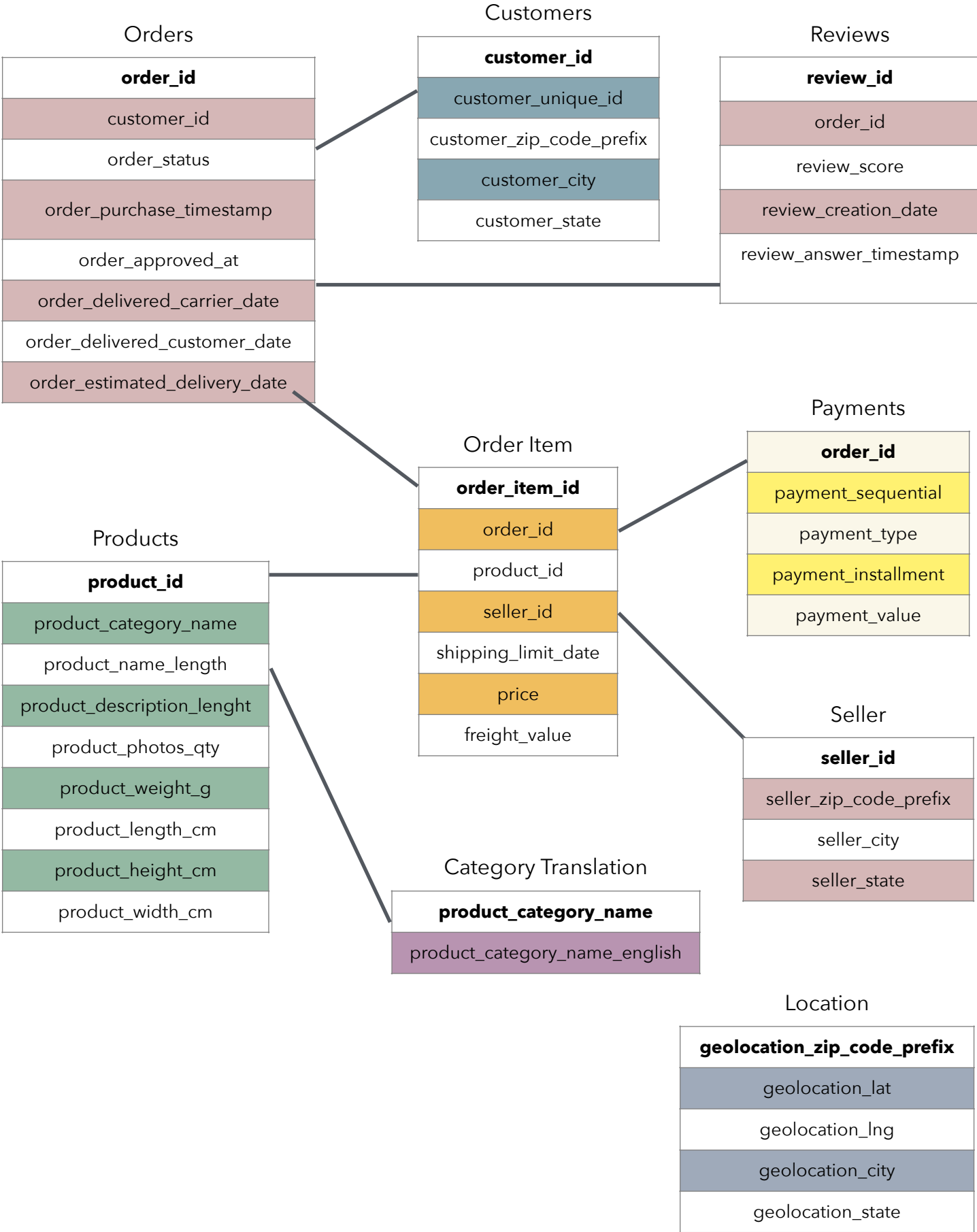


Business Problem Statement

E-commerce businesses generate data across multiple independent systems — orders, payments, products, sellers, and customer feedback — but these datasets are stored in silos, making it impossible to extract unified insights.

This analysis aims to integrate these fragmented datasets into a single master dataset to uncover actionable insights across customer behaviour, sales performance, delivery efficiency, and seller quality.

Dataset Description (Star Schema)



Description

The datasets presented in the table above contain valuable information across multiple dimensions of the e-commerce pipeline. However, in their fragmented state, these datasets are insufficient for deriving meaningful insights in isolation. Therefore, integrating these datasets into a unified master dataset is a necessary prerequisite for conducting comprehensive analysis and extracting actionable business intelligence.

Methodology

The master dataset was constructed through a series of sequential merges across seven independent datasets – orders, customers, order items, products, payments, reviews, and sellers. The resulting unified dataset consolidates all transactional, behavioural, and operational dimensions of the e-commerce pipeline into a single structured table. This integration eliminates data fragmentation and establishes a comprehensive analytical foundation, enabling cross-dimensional analysis across customer behaviour, product performance, seller quality, delivery efficiency, and revenue generation

- **Orders + Customers** This merge establishes a relational link between transactional order records and customer demographic data, forming the foundation for customer-level behavioural analysis and segmentation.
- **Orders + Order Items** By joining order records with their corresponding line items, this merge enables granular analysis at both the order and product level, capturing the composition of each transaction.
- **Order Items + Products** This integration maps each purchased item to its product attributes, facilitating the evaluation of product performance, category demand, and inventory insights.
- **Orders + Payments** Associating payment records with their respective orders allows for a comprehensive examination of revenue flows, payment method distribution, and transaction value analysis.
- **Orders + Reviews** Linking customer feedback to specific orders enables the systematic assessment of customer satisfaction and its relationship to order fulfilment outcomes.
- **Order Items + Sellers** This merge establishes a connection between individual order items and the sellers responsible for fulfilment, enabling seller performance evaluation and quality benchmarking.
- **Products + Category Translation** Integrating the category translation table resolves linguistic inconsistencies by converting Portuguese category labels into English, ensuring dataset interpretability and analytical consistency across a broader audience.

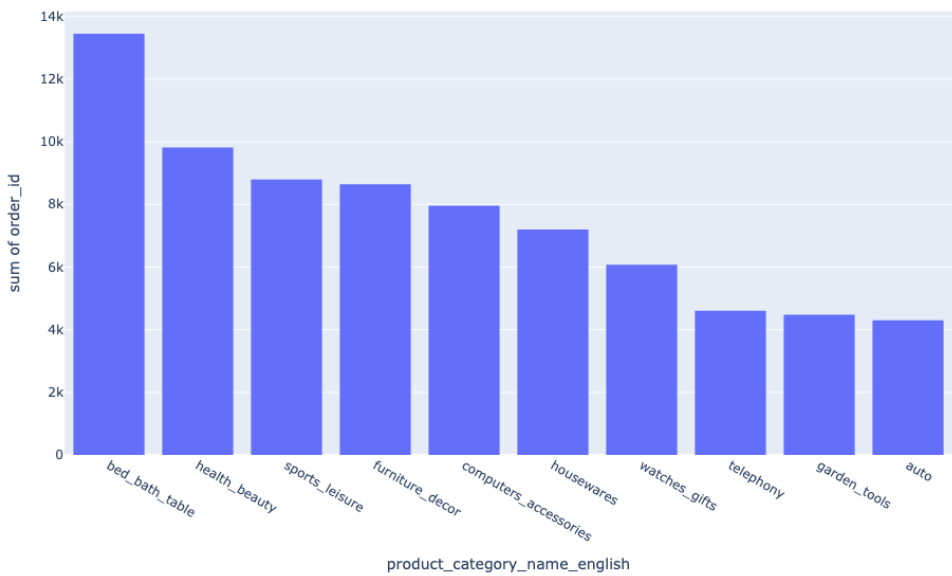
Key Findings

Temporal Trends August and May recorded the highest order volumes, while May also emerged as the most profitable month. Conversely, September demonstrated the weakest financial performance across the observed period.

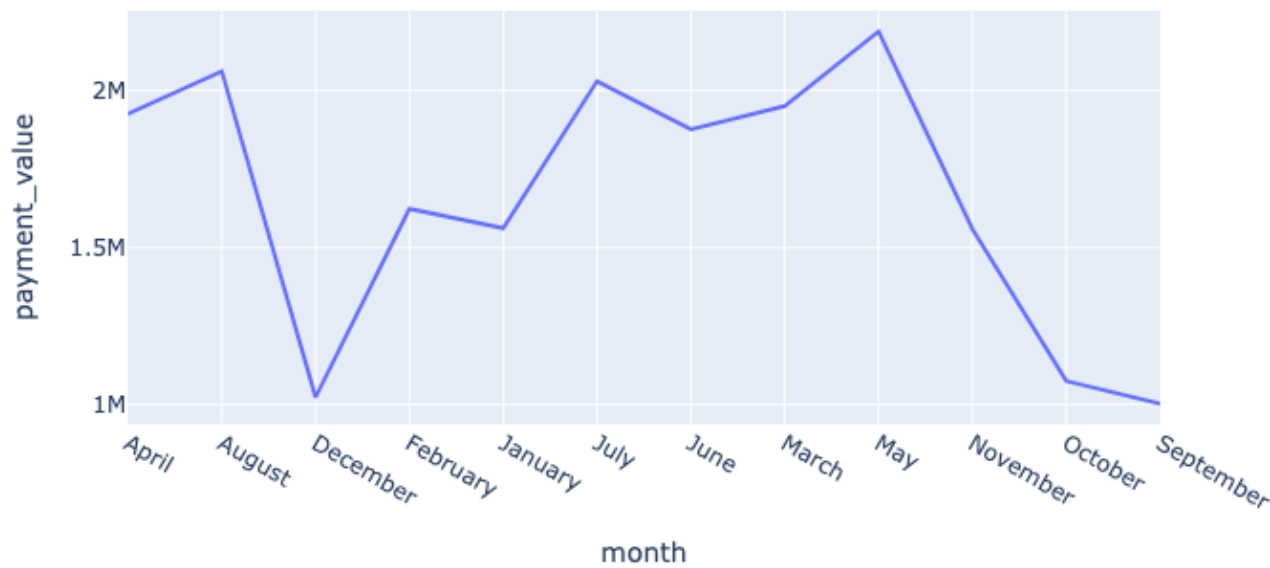
Category Performance The top performing product categories by order volume were Bed & Bath, Health & Beauty, and Computers & Accessories, with Bed & Bath recording its highest concentration of orders in São Paulo.

Geographical Insights São Paulo dominated order volume across all cities, reinforcing its position as the primary commercial hub within the dataset.

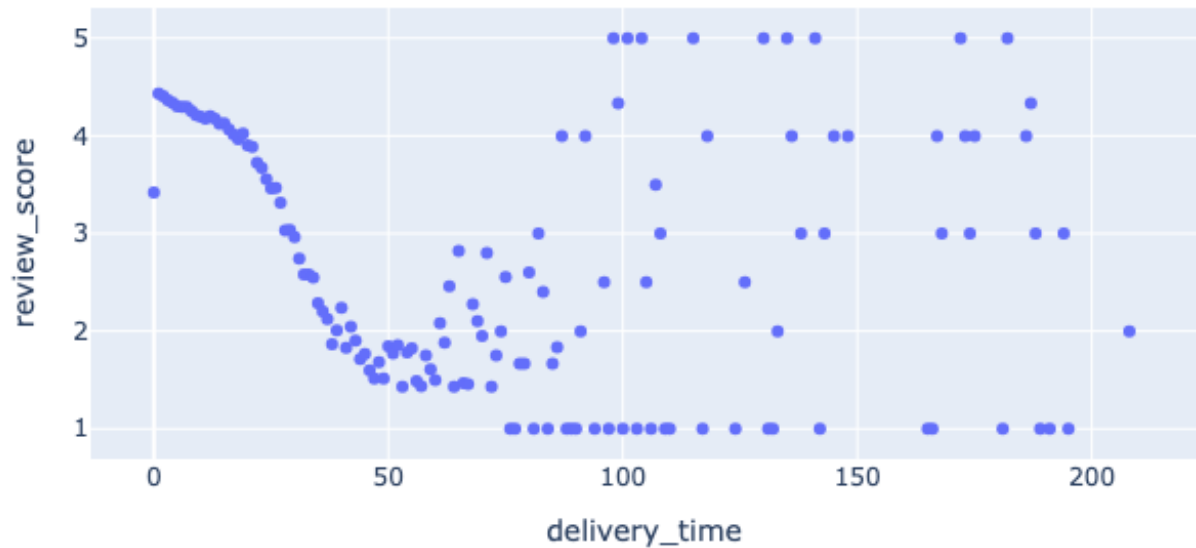
Delivery & Customer Satisfaction Analysis of delivery performance revealed that Iomere achieved the fastest average delivery times, while Santo Antônio recorded the lowest average review scores. Furthermore, the data suggests that while fast delivery is generally associated with higher customer satisfaction, extended delivery times do not consistently correlate with negative reviews, indicating that additional factors influence customer perception beyond logistics alone.



Order Count to Category

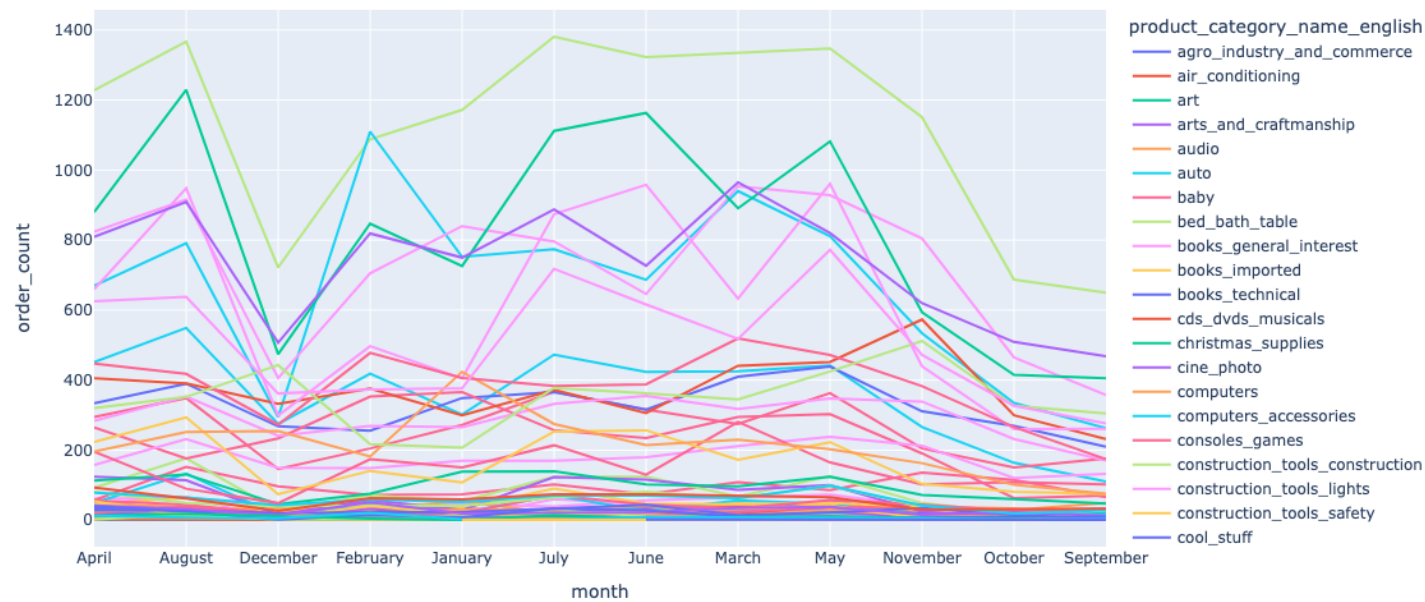


Value to Month



Delivery time to Review Score

Orders by Category per Month



Order by Category per Month

Insights

Seasonal Demand & Revenue August and May consistently record the highest order volumes and revenue, indicating strong seasonal demand during these periods. September demonstrates the weakest financial performance, suggesting a significant drop in consumer activity.

Category Performance Bed & Bath, Health & Beauty, and Computers & Accessories emerge as the dominant product categories by order volume, reflecting sustained consumer demand. Bed & Bath orders are heavily concentrated in São Paulo, highlighting a strong regional preference.

Geographical Distribution São Paulo accounts for the largest share of orders across all cities, indicating a highly concentrated customer base and its significance as the primary commercial hub.

Delivery & Customer Satisfaction Fast delivery times are positively associated with higher review scores. However, poor delivery times do not consistently result in negative reviews, suggesting that product quality and customer service are compensatory factors in shaping customer perception.

Seller Performance Considerable variation in delivery times and review scores across cities and sellers points to inconsistencies in service quality across the network.

Recommendations

Seasonal Demand & Revenue Businesses should increase marketing spend, expand inventory, and launch promotional campaigns ahead of August and May to maximise peak season revenue. Targeted strategies such as discounts and loyalty incentives should be deployed during September to counteract the observed revenue decline.

Category Performance Investment should be directed towards expanding product assortments within the top performing categories. Underperforming categories should be reviewed for potential consolidation or repositioning to optimise catalog efficiency.

Geographical Distribution While São Paulo should continue to be prioritised, businesses should explore targeted expansion strategies in secondary cities to diversify revenue streams and reduce geographical dependency.

Delivery & Customer Satisfaction Investment in last-mile delivery optimisation, particularly in regions with slower delivery times, is strongly recommended. Simultaneously, maintaining high standards in product quality and customer service should be prioritised as key drivers of customer satisfaction beyond logistics alone.

Seller Performance A structured seller performance monitoring framework should be implemented, with clear benchmarks for delivery efficiency and customer satisfaction, to ensure consistent service quality across the seller network.